Quantum Measurements and Gravitational Microstates

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May 16, 2026

Abstract

Thermodynamic approaches to gravity, from Jacobson to Verlinde, relate spacetime dynamics to horizon entropy but leave unspecified the microscopic carriers of individual entropy increments. We identify these with quantum measurement events, realized explicitly as helicity-anticorrelated Bell states of photons sourced by a bilocal driving term in the Rindler setting. The central result is an event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ between the Gibbons–Hawking–York boundary term and the matter action, with the Planck area ℓ_P^2 as the conversion factor: horizon geometry counts measurement events in geometric language. From a single temperature-eliminated Unruh–Landauer bound, standard black hole thermodynamics follows from Newton's laws. These microstates are grounded in a concrete information-theoretic picture of matter: massive bodies are timelike channels sampling internal degrees of freedom at the Bremermann–Nyquist rate mc^2/h, massless quanta are pure carriers between such channels, and horizons are the spacelike surfaces at which handoff events occur, with proper time emerging as accumulated quanta of action along the worldline. This provides a framework consistent with Wheeler's it-from-bit, in which the physical record of each binary measurement outcome is written into matter, and spacetime geometry is a derived property of matter's information accounting.

1 Introduction

Two landmark results establish that gravity and Unruh thermodynamics are deeply intertwined. Jacobson showed that the Einstein equations can be derived from thermodynamic identities applied to local Rindler horizons, with horizon entropy playing the role of the fundamental quantity from which spacetime dynamics emerge [1]. Verlinde extended this perspective, arguing that gravitational forces arise from entropy gradients on holographic screens, recasting gravity as an entropic rather than a fundamental interaction [2]. Both frameworks, however, operate at the level of bulk thermodynamics and leave open a prior question: what physical process underlies each individual entropy increment?

We argue that the answer can be quantum measurement. Each act of information acquisition by a physical observer constitutes a discrete microstate event, and the collective backreaction of such events, together with the causal correlations they induce, give rise to gravitational effects: Any observation is mediated by a physical interaction, information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. Then the equivalence principle makes this gravitational: the local quantum field theory near a black hole horizon is identical to that of a uniformly accelerated observer in flat spacetime, so the Unruh effect and the Hawking effect are the same computation in different coordinates [3, 4].

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We proceed in two complementary way. First, we construct explicit microstates: individual terms of the Unruh thermal ensemble, realized as helicity-anticorrelated Bell states of photons sourced by a bilocal driving term, each corresponding to a single quantum measurement event with a definite energy-momentum transfer. Second, and more fundamentally, we develop an information-theoretic framework that makes this identification natural: massive matter as a timelike channel sampling internal degrees of freedom at the Bremermann–Nyquist rate mc^2/h, massless quanta as pure carriers between such channels, and horizons as the spacelike surfaces at which channel handoffs occur.

The microstate construction is the concrete quantum-mechanical realization of the channel picture, and the channel picture is what makes each microstate inevitable: a measurement event is a handoff between channels, and a handoff has exactly the properties a gravitational microstate must have. The central technical result, $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$, sits precisely at their intersection: it is simultaneously the statement that horizon geometry counts measurement events microstate by microstate, and that the Bekenstein spatial count of Planck-area patches equals the Bremermann temporal count of action quanta, with the Unruh–Landauer relation as the bridge connecting the two directions of counting.

A structuring choice runs throughout the paper: we work in the non-thermal global Minkowski description rather than the thermal Rindler description. The Rindler observer's thermal bath arises from restricting to one wedge and tracing out the other [5, 3, 6]; the Minkowski vacuum is pure, and the individual microstates constructed here are non-thermal. Temperature is not suppressed but relocated: it is a derived quantity in the Rindler frame and absent as a primitive in the Minkowski microstate picture. This vantage point is maintained through Section 6; temperature re-enters in Section 7 not as a retreat to the thermal description but as a demonstration that the non-thermal microstate construction is rich enough to support a fully reversible Carnot cycle at the individual microstate level, with thermodynamic identities emerging as consequences.

1.1 Conceptual Roadmap

Our approach is a refactored version of the Diósi–Penrose objective collapse programme [7, 8], which postulates that gravity induces an objective, observer-independent collapse of the quantum state. In that framework, gravitational self-interaction is taken to define a physical timescale for wavefunction reduction, leading to a dynamical modification of standard quantum mechanics. We instead treat quantum measurement as a subjective process mediated by energy–momentum transfer, and interpret the reduction of the state as physical. The contrast between the two approaches is illustrated in Figure 1. The three solid arrows represent "grounding relations"; the single dashed arrow represents the Diósi–Penrose causal postulate. It is the only relation in the diagram claimed to be objective and observer-independent, and it is the only one we do not adopt in this paper.

In Section 2 we explore a heuristic argument where we eliminate temperature using a combination of Landauer's principle with the Unruh effect and derive standard black-hole thermodynamics area and information laws as a consequence of Newton's laws. We then dive into a spin-1 version of the concrete microstate construction from our previous work [9] which we present in Section 3; this realizes individual terms of the Unruh thermal ensemble via a Minkowski bilocal driving source. In Section 4 we develop a language for discussing emergent spacetime as a property of information bearing matter. Section 5 develops how the Gibbons–Hawking–York (GHY) horizon boundary term [10] counts matter's information as action and models our entangled photons as AS pp-waves that create space at the wedge horizons, following the gravitational shockwave framework of Dray, 't Hooft [11], and others; the central result is that the GHY boundary term equals the matter action event by event, $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$.

Section 6 identifies the newly emergent space with the measurement of a classical bit, and discusses the structural parallel with the ER=EPR correspondence. We circle back

$$\begin{tikzcd}
\text{Gravitational field} \arrow[r, dashed, "\text{Diósi–Penrose}"] & \text{Quantum state reduction} \\
\text{Acceleration} \arrow[u, "\text{equivalence principle}"'] \arrow[r, "\text{energy–momentum transfer}"'] & \text{Observation} \arrow[u, "\text{Bayesian update}"']
\end{tikzcd}$$

Figure 1: Solid arrows denote subjective grounding relations: $A \to B$ means every instance of B is physically realized by an instance of A, for a given observer. The dashed arrow denotes the Diósi–Penrose postulate, where the relation is claimed to hold objectively rather than observer-relatively.

to temperature in Section 7 with a Carnot cycle at the microstate level, connecting the Carnot efficiency bound to the Unruh–Landauer bound as the same inequality viewed from opposite directions. Section 8 develops the effective field theory structure of the construction: the GHY term arises as the effective action for the right wedge after integrating out the left wedge in the path integral, the Planck length emerges as the geometric mean of the two natural scales, and six equivalent conditions characterising EFT validity are shown to coincide at the horizon. We finish with gauge consistency in Appendix 11.

2 Motivating Heuristics

Landauer's principle [12] is conventionally formulated for information erasure. Bennett [13] showed that measurement itself can in principle be performed reversibly; it is the erasure of the measurement record that necessarily costs energy. The bound we derive here is a constraint on the energy of the physical carrier required to imprint a classically distinguishable record under proper acceleration a.

For an accelerated observer, the relevant temperature is the Unruh temperature [3]. Substituting one into the other eliminates temperature entirely, yielding a relation between information acquisition, energy transfer, and acceleration (a version of the black hole thermodynamics first law; equation (3) upcoming). This creates a dictionary between the thermal description natural to a Rindler observer and the non-thermal global Minkowski description [5, 6]: the Minkowski vacuum is pure and non-thermal; restriction to one Rindler wedge produces a mixed thermal state by tracing out the complementary wedge, making thermality a consequence of restricted causal access rather than a fundamental property of the state. The present paper operates in the non-thermal Minkowski description throughout; the thermal Rindler picture is a derived coarse-graining. In this way we will derive a sequence of standard results in black hole thermodynamics as a consequence of Newton's laws, running Verlinde's entropic programme [2] in reverse: where Verlinde recovers Newton's laws from entropy gradients on holographic screens, the present argument recovers black hole thermodynamics from Newton's law of gravitation and the Unruh–Landauer relation.

2.1 The Unruh-Landauer Relation

Landauer's principle bounds the minimum energy of any physical carrier representing ΔH nats of information[1] as distinguishable degrees of freedom in a thermal environment at

[1] We are going to use S for action later so use H here for entropy with units of nats to avoid carrying around a $\ln(2)$.

temperature T:

$$\Delta E \geq k_B T \, \Delta H. \tag{1}$$

The factor $k_B T$ is the thermal energy scale of one distinguishable degree of freedom at temperature T—the equipartition energy of one bit. This is Landauer's own introductory framing: a degree of freedom carries kT of thermal energy, and any physical signal must have at least this energy to be distinguishable from thermal noise [12]. This bound applies to any physical process engaging such a degree of freedom in the thermal environment; the direction of information flow, whether acquisition, erasure, or transmission, does not change the minimum energy required for the carrier to be physically distinguishable. We apply it here as a constraint on the carrier energy: to physically represent ΔH nats in the Rindler vacuum, the carrier must engage at least ΔH nats of thermally distinguishable degrees of freedom. The full thermodynamic accounting of the measurement cycle, where the Landauer erasure cost appears at re-emission, is given in Section 7.

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta H.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a temperature-independent lower bound on the energy of the physical carrier required to register one nat of information under proper acceleration a; acceleration sets the minimum energy scale of the carrier. The full thermodynamic accounting, where the Landauer erasure cost appears correctly, is deferred to Section 7.

This Unruh–Landauer bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $2\pi/\ln 2$:

$$\lambda_{\text{sat}} = \frac{2\pi}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \tag{4}$$

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$, too short to be super-horizon and too long to deposit more than one bit. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8 \text{ m s}^{-2}$, $\lambda_{\text{sat}} \approx 8$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

2.2 Momentum Transfer and Acceleration

In what follows, standard results emerge as consequences of the Unruh-Landauer relation and Newton's laws. The energy transfer ΔE implied by the relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction a *thrust quantum.* For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{5}$$

A measurement event localized to a spatial scale r is bounded by a causal diamond. The temporal extent of this diamond is $\Delta t = 2r/c$, fixed entirely by the causal geometry of the event rather than by any dynamical assumption. We identify the interaction time with this natural causal duration, so that

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2m_d r}. \tag{6}$$

This perspective is closely related in spirit to Erik Verlinde's entropic gravity [2], where forces arise from entropy gradients, but differs in that the present construction grounds each entropy increment in a discrete measurement event with an associated energy-momentum transfer, rather than in a coarse-grained thermodynamic gradient.

2.3 The Schwarzschild Radius

We match the kinematic acceleration of the detector to the gravitational field of the thrust quantum using Newton's laws. By $E = mc^2$, the energy quantum ΔE gravitates with effective mass $\Delta E/c^2$, producing a Newtonian gravitational acceleration

$$a = \frac{G\Delta E}{c^2 r^2} \tag{7}$$

at distance r from the detector. Equating this with the measurement-induced acceleration from equation (6),

$$\frac{\Delta E}{2m_d r} = \frac{G\Delta E}{c^2 r^2}. \tag{8}$$

The energy ΔE of the thrust quantum cancels entirely: the Schwarzschild radius is independent of the energy of the carrier. Solving for r:

$$r = \frac{2Gm_d}{c^2} \equiv r_s, \tag{9}$$

which is precisely the Schwarzschild radius of the detector mass m_d. The cancellation of ΔE is the physical content of the result: whatever energy the thrust quantum carries, the self-consistent horizon scale is set by the detector alone.

We note that the use of Newtonian gravity here is a weak-field approximation, justified when $r \gg r_s$, and that the self-consistency condition $r = r_s$ sits at its boundary of validity. The fact that the exact GR result agrees with this heuristic matching is a consequence shared with Verlinde's entropic gravity [2] and Bekenstein's original argument [14], both of which recover correct relativistic scales from Newtonian reasoning applied at the horizon.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{Gm_d}{r_s^2} = \frac{c^4}{4Gm_d}, \tag{10}$$

in agreement with the standard Schwarzschild result. This is the acceleration that appears in the Unruh temperature associated with the detector's horizon, $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the framework.

2.5 Bekenstein-Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a—justified by the equivalence principle: the local quantum field theory at the black hole horizon is the same as in the Rindler frame [3, 4]—the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \leq \frac{2\pi c}{\hbar\kappa}\Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot c^2 \Delta m_d. \tag{11}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \leq \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{12}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein's original argument [14], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the minimal energy cost of registering information at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$ — which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and c are restored where they aid physical interpretation.

The arguments of Section 2 establish a heuristic relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. Following [9], we invert this logic: we place an interaction in the past, where a source injects non-thermal entangled excitations into the field, and show that the observer's accelerated response can be driven by these source-induced excitations. The thermal ensemble then emerges as a coarse-grained aggregate of such microstates, rather than being a primitive property of the vacuum.

Each microstate, equation (23) upcoming, is individually non-thermal in the sense that it carries a definite Rindler frequency ω, the frequency natural to $\mathcal{R}$ and $\mathcal{L}$, which differs from the Minkowski frequency seen by $\mathcal{P}$ due to the position-dependent redshift of the Rindler geometry. Thermality is not a property of any single microstate but of the ensemble; it emerges when one asks only which wedge a detection occurred in, discarding the specific microstate identity, and is a consequence of coarse-graining rather than a premise of the construction.

The thermal Unruh distribution is the correct description for an observer with no knowledge of the source of their acceleration: without access to $\mathcal{P}$'s emission history, all possible microstates appear equally likely, and both the Planck distribution over ω and the mixed density matrix obtained by tracing out the complementary wedge reflect the same epistemic situation. The bilocal construction breaks this ignorance explicitly. Because $\mathcal{P}$ emits at a definite frequency ω, a measurement—a PVM projection onto a definite helicity outcome—resolves both coarse-grainings simultaneously: after the collapse there is no ω-averaging and no partial-trace ambiguity, and the observer is at a specific point in microstate space within the non-thermal Minkowski description. In this sense measurement is access to the Minkowski description, and the apparent thermal bath is replaced by a definite momentum transfer—a thrust—sourced by $\mathcal{P}$. This reinterprets a portion of the Unruh response as physically sourced thrust rather than thermal fluctuation [9], and it is the concrete realisation of the Bayesian update arrow in Figure 1: state reduction is not imposed from outside but follows from the physical interaction at the measurement event.

We now upgrade this construction from a spin-0 scalar field in our original source [9] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer. The Bell state is the natural completion of the CPT[2] symmetry between wedges, which by the Bisognano–Wichmann theorem [6] is implemented by the modular conjugation operator J mapping the right wedge algebra to the left wedge algebra, and as we discuss in Section 5, the photons imprint a definite geometric signature onto the horizon.

3.1 A Source and Two Observers

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ - *Right*, $\mathcal{L}$ - *Left*, and $\mathcal{P}$ - *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of

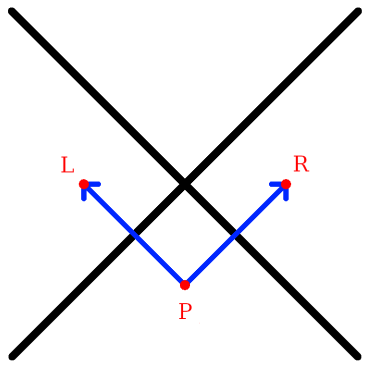

Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ into the right Rindler wedge and $\mathcal{L}$ into the left Rindler wedge. The two null directions of propagation define the z^+ and z^- axes of the double-null coordinate system.

spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The momenta are perfectly balanced, so the source exhibits no net back-reaction in this minimal model. The two photons propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them respectively, each registering a helicity eigenvalue and acquiring one classical bit of information.

The three-party structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$ is a tripartite entanglement: neither $\mathcal{R}$ nor $\mathcal{L}$ alone holds sufficient information to reconstruct the emission event at $\mathcal{P}$, but their two complementary bits together do, making $\mathcal{P}$ the logical AND of the pair and providing a natural microstate-level realization of interior reconstruction.

3.2 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$,

[2]PCT in Bisognano–Wichmann.

with equal and opposite spatial momenta. These two null directions are precisely the z^+ and z^- axes of double-null coordinates

$$z^{\pm} = \frac{1}{\sqrt{2}}(t \pm z), \tag{13}$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \qquad A_-(z) = 0, \tag{14}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the CPT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$; consistency with Maxwell's equations and the absence of over-determination are verified in Appendix 11.

The lack of manifest four-dimensional covariance in equation (14) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k^\mu_R, k^\mu_L\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice.

3.3 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$J^{ij}(x, y) = g \left(\epsilon^{i*}_{(-1)} \, \epsilon^{j}_{(+1)} + \epsilon^{i*}_{(+1)} \, \epsilon^{j}_{(-1)} \right) f_L(x) \, f_R(y), \tag{15}$$

where λ is a coupling constant, f_L and f_R are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{16}$$

The two terms in (15) are related by the CPT transformation in the sense of Bisognano–Wichmann [6], which maps the right wedge algebra to the left wedge algebra and simultaneously flips $\lambda \to -\lambda$. The symmetrization is therefore not an ad hoc choice but the unique assignment consistent with CPT symmetry and angular momentum conservation at the emission vertex; the total angular momentum vanishes in both terms.

The sourced Lagrangian reads

$$\begin{aligned} \mathcal{L}_{\text{sourced}} &= \mathcal{L}_{\text{free}} + \tfrac{1}{2} g \int J^{ij}(x, y) \; A^+_i(x) \, A^+_j(y) \; d^4x \, d^4y \; + \; \text{h.c.} \\ &= \mathcal{L}_{\text{free}} + \tfrac{1}{2} g \Big(a^{L\dagger}_{-k,(-1)} \, a^{R\dagger}_{k,(+1)} \; + \; a^{L\dagger}_{-k,(+1)} \, a^{R\dagger}_{k,(-1)} \; + \; \text{h.c.} \Big), \end{aligned} \tag{17}$$

where the two creation operator pairs correspond to the two CPT-related helicity assignments. The interaction term in (17) is a two-mode squeezing interaction at frequency ω, now resolved by helicity and symmetrized over both assignments: it is the direct vector-field analogue of the scalar squeezing interaction [15], with the CPT symmetry between wedges made explicit. The photon pair carries total angular momentum zero in each term, consistent with emission from a scalar source at $\mathcal{P}$.

Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$, identifying the thrust quantum of Section 2 with a concrete physical object: the minimum actionable event is precisely a single photon. Its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

The Unruh–Landauer bound is saturated not at the level of an individual photon but at the level of the source rate. The acceleration a does not fix the frequency ω of each emitted quantum; rather, it determines the rate $\dot{N}$ at which the bilocal source must produce entangled pairs in order to fully[3] implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux. Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome — one bit in the sense of one binary alternative, carrying $\ln 2$ nats — and these are paired by the construction independently of a. The Landauer bound then operates as a consistency condition on the ensemble: the minimum energy cost per bit acquired, summed over the rate of acquisition, equals the power required to sustain the acceleration.

3.4 Polarization Bogoliubov Transformation

We now show that the polarization labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A^i_\lambda(z) = \int_0^\infty \frac{d\omega}{2\pi}\,\frac{1}{\sqrt{2\omega}}\,\epsilon^i_{(\lambda)}\left[b^R_{\omega,\lambda}\,u^R_\omega(z) + b^{R\dagger}_{\omega,\lambda}\,u^{R*}_\omega(z)\right], \tag{18}$$

where $u^R_\omega(z)$ are the right-wedge Rindler modes at Rindler frequency ω, and $b^R_{\omega,\lambda}$, $b^{R\dagger}_{\omega,\lambda}$ are the corresponding annihilation and creation operators satisfying

$$\left[b^R_{\omega,\lambda},\, b^{R\dagger}_{\omega',\lambda'}\right] = \delta(\omega-\omega')\,\delta_{\lambda\lambda'}. \tag{19}$$

An identical expansion holds in the left wedge with $R \to L$ and $\lambda \to -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes the same form in each sector. For helicity λ at Rindler frequency ω, the Unruh modes are

$$c_{\omega,\lambda} = \cosh\theta_\omega\; b^R_{\omega,\lambda} - \sinh\theta_\omega\; b^{L\dagger}_{\omega,-\lambda}, \tag{20}$$

where $\tanh\theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration a, and the CPT symmetry between wedges is reflected in the superposition over both helicity assignments in the Bell state. The transformation (20) is the standard Bogoliubov relation for a massless field [3, 16], now carrying a helicity label throughout. The Minkowski vacuum thermofield double state, in the Rindler basis reads, for each helicity sector,

$$|0\rangle^{(\lambda)}_M = \frac{1}{\cosh\theta_\omega}\sum_{n=0}^{\infty}\tanh^n\theta_\omega\;|n_{\omega,\lambda}\rangle_R\,|n_{\omega,-\lambda}\rangle_L, \tag{21}$$

and the full vacuum is the product over all ω and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda}|0\rangle^{(\lambda)}_M. \tag{22}$$

[3] The bilocal source may be only part of the acceleration, so $\dot{N}$ is actually a lower bound.

Equation (21) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source selects a single term from this sum at fixed ω and fixed helicity pair $(\lambda, -\lambda) = (+1, -1)$, yielding the single-microstate

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} \left(|1_{\omega,+1}\rangle_R \, |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R \, |1_{\omega,+1}\rangle_L\right) + \mathcal{O}(\text{higher order}). \tag{23}$$

$\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes are undetermined until measurement, at which point they are perfectly anticorrelated by the Bell state structure: each observer acquires one classical bit, and the two bits are complements of one another.

The key structural result is that the helicity label λ is conserved under the Bogoliubov transformation: it enters equation (20) as a spectator index and exits unchanged on the Unruh operators $c_{\omega,\lambda}$. Thermality, when it arises from coarse-graining over the ensemble of microstates (23), is therefore helicity-resolved: each helicity sector thermalises independently, and the anticorrelation between wedges is preserved at every level of the construction.

4 Matter as a Substrate

Quantum mechanics already endows every massive particle with a natural channel structure. A particle of mass m accumulates phase at the Compton frequency $f_C = mc^2/h$; the Bohr–Sommerfeld condition $\oint p\,dq = nh$ expresses this as the accrual of phase in integer units of h, so that physical evolution is discrete at the scale set by the action. Bremermann's computational limit [17] recovers the same rate mc^2/h as the maximum number of distinguishable operations per second for any system of energy mc^2, and Lloyd [18] shows this bound is saturated by quantum systems. The Compton frequency is therefore simultaneously a clock rate, a sampling rate, and a computational limit—three facets of the single fact that matter's state space has a resolution set by $\hbar$. Proper time is accumulated action along a worldline, an emergent quantity in the sense of Page and Wootters [19, 20]: time arises from physical correlations between systems rather than from a background parameter, with the Bremermann rate setting the resolution at which those correlations are sampled. This section develops the channel picture and derives three results that Section 5 realises geometrically: the proportionality of the Bremermann and Bekenstein bounds (Section 4.2), a formal definition of the handoff event through which space is generated at horizons (Section 4.3), and the saturation of channel capacity precisely at the Rindler horizon (Section 4.4).

This viewpoint sidesteps a prior question about quantum gravity: how can one quantize gravity without directly quantizing spacetime or its curvature degrees of freedom? Common approaches posit Planck-scale cells [21] or discrete spacetime events [22], but Lorentz invariance is recovered only statistically in such constructions rather than at the level of individual configurations [23]. We instead derive discreteness from matter: space and time are properties of matter itself, defined only up to the resolution set by the action $S/\hbar$, which is Lorentz invariant by construction.

4.1 Matter as a Timelike Channel

A massive particle carries phase along its worldline at a rate set by its rest energy. The Compton frequency $f = mc^2/h$ is the natural clock rate of a mass m, and the Bohr–Sommerfeld quantization condition $\oint p\,dq = nh$ is its coarse-grained expression: it is the correct effective description at energy scales below mc^2, where the sub-Compton phase structure is unresolved and action accumulates in integer units of h. This is not a limitation but a feature — the condition is technically precise in the Heisenberg sense, valid exactly within the EFT window defined by the Compton scale as UV cutoff and the horizon distance as IR cutoff (Section 8). Bremermann's computational limit [17] independently recovers the same rate as the maximum clock rate of any physical system with energy E, and the two agree because $\hbar$ is the minimal unit of physical evolution at that scale.

Each quantum corresponds to one sample[4] of the phase $e^{iS/\hbar}$ in the path integral. The matter substrate gives rise to a channel capacity in the Shannon sense: for a system of mass m with internal Hilbert space dimension d, the maximum rate of information flow is

$$C = \underbrace{\frac{mc^2}{\hbar}}_{\text{bandwidth}} \times \underbrace{\log_2(d)}_{\text{dynamic range}} \qquad \text{(bits per second)}, \tag{24}$$

where the bandwidth is the Bremermann sampling rate and the dynamic range is the information content per sample. Both quantities are extensive under composition:

$$\frac{mc^2}{\hbar} = \frac{(m_1+m_2)c^2}{\hbar} = \frac{m_1c^2}{\hbar} + \frac{m_2c^2}{\hbar}, \tag{25}$$

$$\log_2(d) = \log_2(d_1d_2) = \log_2(d_1) + \log_2(d_2), \tag{26}$$

consistent with their interpretation as properties of matter that compose naturally under union. The signal being sampled at the Bremermann rate is the internal quantum state of the particle — its spin, helicity, and other quantum numbers — and the dynamic range is $\log_2$ of the dimension of the internal Hilbert space. The channel capacity (24) sets the rate at which matter can accumulate and process information; Section 4.4 shows that this rate saturates at the Rindler horizon, making the horizon the natural IR boundary of the matter channel and anticipating the EFT breakdown of Section 8.

4.2 The Horizon as a Spacelike Channel

The three fundamental constants c, $\hbar$, and G play precise roles in the channel picture. The speed of light c sets the causal propagation speed of signals. The reduced Planck constant $\hbar$ sets the resolution of the timelike channel via the Compton frequency $mc^2/\hbar$. The gravitational constant G, through the Schwarzschild radius $r_s = 2Gm/c^2$, sets the maximum energy storable in a spatial region without gravitational collapse. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial region per independent quantum of action.

The horizon enters the channel picture as the surface at which the matter-memory on one side becomes causally disconnected from the matter-memory on the other. The photon passes through the horizon channel and deposits into the matter channel on the other side; the handoff is the central event of the construction.

The Bekenstein and Bremermann bounds are not merely analogous: they count the same physical event from orthogonal directions, related by the Unruh–Landauer relation of Section 2. To see this explicitly, work in natural units $\hbar = c = 1$ and express frequencies in units of the Rindler acceleration a, so that $\Omega \equiv \omega/a$ is dimensionless. For a single photon of dimensionless Rindler frequency Ω crossing the horizon, the Bremermann count is the number of new resolution elements added to the receiving timelike channel per Rindler period $\tau = 2\pi/a$:

$$\Delta f \cdot \tau = \frac{\omega}{2\pi} \cdot \frac{2\pi}{a} = \Omega. \tag{27}$$

The Bekenstein count of the same event is the spatial entropy increment. By the Raychaudhuri calculation of Section 5, the horizon area increments by $\Delta A = 8\pi G\omega = 8\pi Ga\Omega$,

[4] We use the information theoretic term *sample* throughout to denote a single zero crossing of the phase $e^{iS/\hbar}$ in the path integral. The term *event* or *tick* may be substituted throughout. It is defined only up to the Heisenberg resolution.

giving

$$\Delta H_{\text{Bek}} = \frac{\Delta A}{4\ell_P^2} = \frac{8\pi G a\Omega}{4G} = 2\pi\Omega. \tag{28}$$

The two counts satisfy

$$\boxed{\Delta H_{\text{Bek}} = 2\pi \cdot (\Delta f \cdot \tau),} \tag{29}$$

with the factor 2π being the Rindler normalisation—the same factor appearing in the Unruh temperature $T = \hbar a/2\pi k_B c$ and tracing to the analytic continuation between Minkowski and Rindler coordinates. At the fundamental Rindler mode $\Omega = 1$ (one cycle per Rindler period), the Bremermann count gives one new resolution element and the Bekenstein count gives $\Delta H = 2\pi$ nats, consistent with the direct calculation of Section 5. The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ of Section 5 is the geometric proof of (29): counting Planck-area patches spatially and counting action quanta temporally give the same result up to the universal Rindler factor 2π.

4.3 The Handoff Event

The photon crossing the horizon is the handoff between the timelike matter channel and the spacelike horizon channel. We now define this object precisely, so that Section 5 can identify it with a specific geometric structure rather than deriving a coincidental equality.

Definition (Handoff Event). A *handoff event* at a horizon surface $\mathcal{H}$ consists of:

1. a massless quantum of Rindler frequency ω and helicity $\lambda = \pm 1$ transferring one quantum of action $\hbar$ across $\mathcal{H}$;
2. the receiving timelike channel incrementing its Bremermann sampling rate by $\Delta f = \omega/2\pi$;
3. the helicity outcome λ recorded as one classical bit in the receiving channel's matter-memory; and
4. the horizon area incrementing by $\Delta A = 8\pi G\hbar\omega$.

Property (iv) follows from the Raychaudhuri equation applied to the null congruence at $\mathcal{H}$ and is derived in Section 5; it is listed here to make the geometric content of the handoff explicit.

Each handoff event has exactly two irreducible constituents: one quantum of action $\hbar$ written into matter-memory, incrementing the Bremermann sampling rate, and one bit of classical information, the helicity outcome. These are not independent: the action quantum $\hbar$ and the helicity $\lambda = \pm 1$ are two aspects of the same physical object, the energy $\hbar\omega$ and angular momentum $\pm\hbar$ of a single photon. With this definition in hand, the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ of Section 5 becomes a counting statement: the GHY boundary term sums $\hbar\omega$ over handoff events because that is what the term geometrically records, one contribution per event.

4.4 Channel Capacity Saturation at the Horizon

The saturation wavelength $\lambda_{\text{sat}} = (2\pi/\ln 2)(c^2/a)$ of Section 2 acquires a precise meaning in the channel picture. A photon of frequency ω absorbed by a detector increments its Bremermann sampling rate by $\Delta f = \omega/2\pi$. Over one Rindler period $\tau = 2\pi c/a$, the number of new resolution elements added to the timelike channel is

$$\Delta f \cdot \tau = \frac{\omega}{2\pi} \cdot \frac{2\pi c}{a} = \frac{\omega c}{a}. \tag{30}$$

The Landauer saturation frequency follows from $\lambda_{\text{sat}} = 2\pi c/\omega_{\text{sat}}$:

$$\omega_{\text{sat}} = \frac{a \ln 2}{c}. \tag{31}$$

Substituting into (30):

$$\Delta f \cdot \tau|_{\omega=\omega_{\rm sat}} = \frac{\omega_{\rm sat} c}{a} = \ln 2. \tag{32}$$

The Landauer-saturating photon adds exactly $\ln 2$ nats — one bit — of Bremermann channel capacity per Rindler period. A photon below saturation ($\omega < \omega_{\rm sat}$) adds less than one bit per period and cannot constitute a complete measurement record; a photon above saturation adds more than one bit and deposits more than the minimum cost of a handoff event. The horizon is therefore the surface at which the minimum energy cost of information acquisition (Landauer) and the minimum time cost of processing it (Bremermann) coincide in a single-bit handoff. This saturation condition is also the natural boundary of EFT validity: the six equivalent conditions of Proposition 1 all reduce to $\omega = \omega_{\rm sat}$ at the horizon, so the EFT breaks down exactly where the channel saturates.

4.5 Types of Particles and Interactions

The distinction between massless and massive degrees of freedom is operational in the channel picture: massless quanta transmit information between matter substrates as handoff carriers, while massive systems retain it as matter-memory. A massless particle has no rest frame, no proper time, and carries a single quantum of action between emission and absorption; it does not sample and therefore carries no persistent memory. Its role is precisely that of a pure information carrier effecting a handoff between the timelike channel of one massive body and that of another.

A single photon is the minimal carrier because it contributes exactly one of each:

- **One quantum of action $\hbar$**: the physical transfer, deposited as energy $\hbar\omega$ into the detector, incrementing its Bremermann sample rate by $\Delta f = \hbar\omega/h$.
- **One bit of polarization**: the informational outcome, the classical record of which helicity $\lambda = +1$ or $\lambda = -1$ was absorbed, persisting in the detector's state until erased at a minimum cost governed by Landauer's principle.

These two constituents are not independent: the action quantum $\hbar$ and the helicity $\lambda = \pm 1$ are two aspects of the same object, the energy $\hbar\omega$ and angular momentum $\pm\hbar$ of a single photon.

From a given observer's perspective, interactions fall into three classes. **Absorption** produces a persistent classical record: a quantum of action is deposited and information is written. **Emission** is the reverse: stored information is released and the record is lost. **Reflection** is absorption followed by emission; whether it constitutes an information-bearing interaction depends on whether a persistent record survives. These categories are observer-relative in the same sense as quantum state reduction: the same physical process may be information-bearing for one observer and external for another.

4.6 The Zig-Zag Mechanism as Proof of Concept

The channel picture is not an ad hoc construction: the handoff event of Section 4.3 is already realised concretely in Penrose's zig-zag interpretation of the Dirac equation [24], confirming that the Bremermann rate mc^2/h is the Higgs interaction rate rather than a heuristic. The Lorentz algebra decomposes as

$$\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2), \tag{33}$$

where the two $\mathfrak{su}(2)$ factors correspond to opposite helicities [25]. A Dirac 4-spinor decomposes into two Weyl 2-spinors of opposite helicity and equal and opposite momenta, interacting at regular intervals via the Higgs field at the Compton frequency.

The structural parallel with the bilocal construction and the handoff definition is exact: the two Weyl spinors correspond to the two photons emitted by $\mathcal{P}$; the Higgs interaction

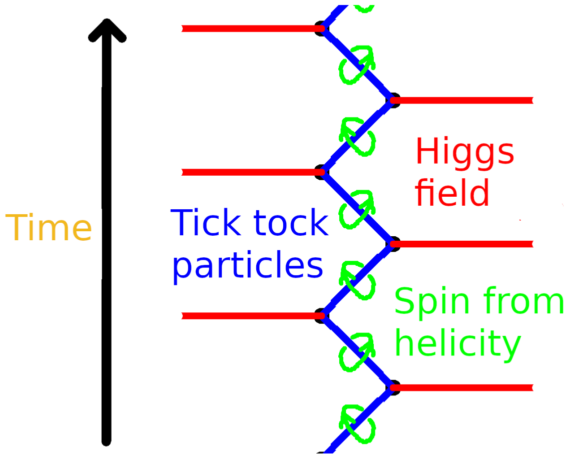

Figure 3: Adapted from [24]. Penrose's interpretation of the Dirac equation and the Higgs mechanism for an electron.

corresponds to the absorption events at $\mathcal{R}$ and $\mathcal{L}$; between interactions each spinor is massless and moves at c, a pure carrier. Each Higgs interaction is a handoff event in the sense of Section 4.3: it deposits one quantum of action $\hbar$, produces one sample of proper time, records one helicity outcome, and increments the matter-memory of the particle by one Bremermann sample. Mass emerges from the regularity of these interactions: a particle is massive because it undergoes repeated Higgs interactions at the Compton frequency, and its mass is proportional to that frequency via $E = \hbar\omega$. Proper time along a worldline is then a foliation by discrete handoff events, each separating the causal past from the causal future of that interaction—the one-dimensional analogue of the Rindler horizon, and matching the holographic screen picture of Verlinde [2]. This is illustrated in Figure **??**.

5 pp-Wave Geometry and Action

WLOG we introduce coordinates for the photon traveling from $\mathcal{P}$ $(-1, 0)$ to $\mathcal{R}$ $(0, 1)$. Let $u = t - z + 1$ and $v = t + z + 1$, where the photon propagates along the null plane $u = 0$, z is the longitudinal direction, and (x, y) are transverse coordinates. Then $(u, v) = (0, 0)$ is $\mathcal{P}$'s location, $(u, v) = (0, 2)$ is $\mathcal{R}$'s location, and $(u, v) = (0, 1)$ is the intersection of the photon with the past right Rinder horizon. The situtation is reversed for the left moving photon, we swap u and v to describe photon moving from $\mathcal{P}$ to $\mathcal{L}$.

5.1 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{34}$$

This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{35}$$

directed toward the axis. Test particles are pulled toward the centre of the wave, regardless of their initial transverse separation.

5.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $-v$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 4.

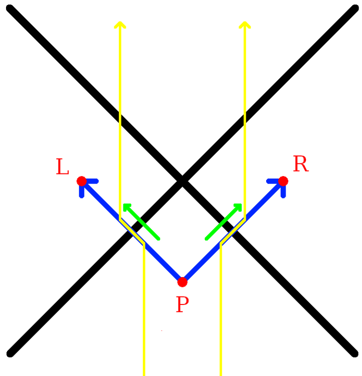

Figure 4: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the creation of space along the null lines originating from $\mathcal{P}$.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse (x, y) plane is identified with the horizon area elements $dA = dx\, dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the v direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the u direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

5.3 Transverse Focusing Increments Horizon Area

The transverse focusing of the AS pp-wave produces a calculable increment in horizon area. We derive this using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1], but applied here to a single microstate rather than a thermodynamic average.

For a null congruence with generators k^μ and expansion θ, the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\tfrac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G\, T_{\mu\nu}k^\mu k^\nu. \tag{36}$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta^\mu_u$, the area change is

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp. \tag{37}$$

In a strict AS construction, the source is a point particle and $T_{\mu\nu}$ is distributional, concentrated on the axis. The present construction provides a natural regularization: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$. The electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane by energy conservation, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale c/ω.

Substituting into equation (37) gives

$$\boxed{\Delta A = 8\pi G\,\hbar\omega = 8\pi G\,\Delta E.} \tag{38}$$

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages.

Applying the Bekenstein–Hawking relation as a heuristic, valid for Schwarzschild but taken here as an ansatz for the Rindler case, and substituting $\Delta E = \hbar\omega$, yields a corresponding entropy increment

$$\Delta H = \frac{\Delta A}{4\ell_P^2} = \frac{2\pi\,\Delta E}{\hbar\omega} = 2\pi, \tag{39}$$

in natural units, consistent with one Planck-area patch per detection event.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each AS pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry. Hawking, Perry, and Strominger [26] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [27] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a BMS_3 algebra whose central charge determines the Bekenstein–Hawking entropy, but his programme identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process: each measurement event produces one AS pp-wave, which deposits one supertranslation at the Rindler horizon, exciting one mode of the BMS_3 algebra. The entropy count is consistent with Carlip's result, the microstates being counted are the measurement events themselves.

5.4 Action and the Gravitational Boundary

Each absorption sample deposits one quantum of action $\hbar$ into the detector's matter-memory. The total action accumulated across all measurement events is therefore

$$S_{\text{matter}} = \sum_i \hbar\omega_i, \tag{40}$$

where the sum is over all absorption samples, each contributing energy $\hbar\omega_i$ to the detector mass. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (38), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G\hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G\, S_{\text{matter}}. \tag{41}$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [10] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}\big|_{\mathcal{H}} = \frac{c^4}{8\pi G}\int_{\mathcal{H}} \theta\, d^2x_\perp\, du = \frac{c^4}{8\pi G}\sum_i \Delta A_i = \frac{c^4}{8\pi G}\cdot 8\pi G \sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \tag{42}$$

The matter action and the gravitational boundary action are therefore equal, event by event, on the stationary Rindler horizon. This is not a bulk result: the bulk Einstein–Hilbert action vanishes on-shell, and it is the boundary term that carries the physical content. The microstate construction identifies what that boundary content is: each contribution to S_{GHY} corresponds to one quantum of action deposited into matter-memory by one measurement event.

The equality $S_{GHY}\big|_{\mathcal{H}} = S_{\text{matter}}$ invites a stronger reading. The bulk Einstein–Hilbert action vanishes on-shell, so it is the boundary term alone that carries the gravitational degrees of freedom. If the boundary term counts matter action quanta event by event, as equation (42) shows, then the geometry at the horizon is not an independent quantity that happens to equal the matter action: it *is* the matter action, expressed in geometric language. The Planck area $\ell_P^2 = \hbar G/c^3$ is the conversion factor between the two descriptions, relating one quantum of action $\hbar$ to one Planck-area patch on the horizon. In this reading the metric near the horizon is not a smooth field that matter happens to perturb; it is a coarse-grained count of the action quanta that have passed through the horizon surface, with the Bekenstein–Hawking area law as the statement that these two counts agree. Whether this identification extends from the boundary into the bulk, that is, whether the Ricci scalar R can be understood as an action-quanta density with ℓ_P^2 as the conversion factor, is a question the present construction motivates but does not settle, and we leave it for future work.

This result is consistent with and extends programmes of Jacobson [1] and Padmanabhan [28], which argue that the gravitational boundary term rather than the bulk Einstein–Hilbert action is the fundamental gravitational degree of freedom. Those programmes establish this at the level of thermodynamic averages. The present construction realizes the same identification at the level of individual microstates, with each boundary contribution sourced by a specific quantum measurement event rather than a bulk average. The standard result emerges from summing over the ensemble; the microstate picture supplies what the thermodynamic programmes leave open: the physical process underlying each boundary contribution.

6 Entanglement as Spatial Information

The bilocal source prepares an anticorrelated Bell state (23). We consider the simplest case: both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and both obtain a definite outcome, which we label H_+ or H_-. In this case their outcomes are perfectly anticorrelated by the Bell state structure, independently of which basis is chosen. The question

"who observed H_+?"

then has a binary answer: $\mathcal{L}$ (left wedge) or $\mathcal{R}$ (right wedge). This is a *spatial bit*, a bit that encodes not a polarization state but a location in space.

Each measurement event has exactly two irreducible constituents: one quantum of action $\hbar$ written persistently into the detector's matter-memory as a series of samples, and one bit of polarization registered as a classical outcome. The entanglement between $\mathcal{R}$ and $\mathcal{L}$ is consumed in producing these two records, an increase in sampling rate and one bit at each wedge. What remains after measurement is not the entanglement itself but its classical residue: the perfect anticorrelation between the two bits, and the correlated pair of samples written into the matter on each side of the horizon. The spatial bit is precisely the polarization bit now read as a location rather than a spin state: the same bit that records *what* was observed also records *where*, because in this construction the two wedges carry opposite helicities by design. This double duty is forced: the anticorrelated helicity assignment of the bilocal source makes the two wedges distinguishable by construction, so the spatial bit is already latent in the source geometry before any measurement occurs.

The information is carried by the matter of the detectors, not by the horizon. As established in Section 4, the horizon is a boundary that makes the left/right distinction sharp; it is the matter on each side that accumulates the samples and holds the classical record. $\mathcal{R}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization H_+/H_-. $\mathcal{L}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization H_-/H_+. The two records are perfectly complementary by the anticorrelated structure of the bilocal source, and their complementarity is encoded in matter and is reflected in geometry.

6.1 Emergent Space and the Thermofield Double

The Unruh vacuum is the thermofield double state of the left and right Rindler wedges [29]: equation (21) is precisely this state, with the helicity label λ carried through. The bilocal source selects one term from this sum at fixed ω and fixed helicity pair, yielding the single microstate (23). The correlated AS longitudinal shifts in opposite null directions are the geometric output of this microstate: a linearized perturbation of flat Minkowski space that displaces the two wedges away from their respective horizons, creating new coordinate space between them. This is structurally similar to the non-traversable ER bridge: the two wedges are connected through the past source $\mathcal{P}$ but no signal can pass directly from $\mathcal{L}$ to $\mathcal{R}$, matching the non-traversability of the original Einstein–Rosen bridge and the no-communication theorem on the EPR side. We do not claim a full topological identification; whether the construction makes contact with the ER=EPR programme in a flat-space limit is left for future work.

7 Carnot Cycle from Unruh–Landauer

The Carnot cycle developed here is formulated at the level of individual non-thermal microstates, and this is the point. Reversibility is a feature of the microstate description: in a coarse-grained thermal treatment individual microstates are inaccessible and the cycle could not be run explicitly. Temperature appears in this section not as a primitive imported from the Rindler description but as a derived label on each leg of the cycle, consistent with the non-thermal Minkowski vantage point maintained throughout. The Carnot efficiency bound and the Unruh–Landauer bound will emerge as the same inequality viewed from opposite sides of this microstate-level construction, closing the loop opened in Section 2.

We consider in this section the case where $\mathcal{L}$ and $\mathcal{R}$ are moving mirrors being accelerated with an outward thrust from our past bilocal source $\mathcal{P}$ emission, see Figure 5. We complete our bilocal microstate absorption example by reflecting and emitting the photons off of the mirrors.

This reflects the energy back to the future sink $\mathcal{F}$, located at $(t, z) = (1, 0)$, which is the CPT image of $\mathcal{P}$ under the combined time-reversal and parity operation that swaps past and future wedges. $\mathcal{F}$ plays the role of a final absorber, receiving the reflected photons and resetting the field state; its existence is required by the reversibility of the ideal mirror cycle and by the CPT symmetry of the bilocal construction. The reflection is reversible in the sense that the mirrors do not retain a persistent copy of the information that they temporarily store.

The mirrors are ideal, in the sense that they do not retain a persistent copy of the information that they temporarily store; they forget and emit. It is this reversibility that allows us to construct a example of a Carnot cycle at the level of microstates. The working system is the photon–mirror interaction degree of freedom; the Rindler wedge defines the thermal reservoirs, and the mirror trajectory defines the work parameter.

Consider the point of view of the right Rindler observer $\mathcal{R}$ who sees the event horizon as a thermal object. The Rindler horizon plays the role of the hot reservoir at the Unruh

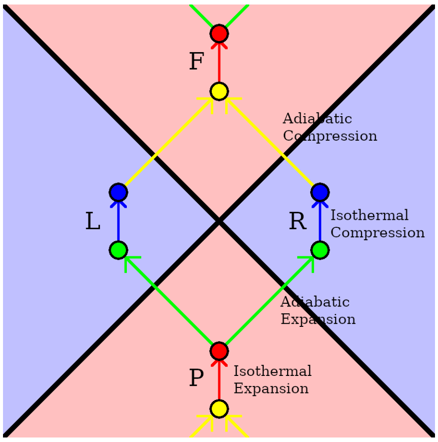

Figure 5: Carnot Cycle: Past ($\mathcal{P}$) bilocal source emits entangled pair with one leg hitting a moving mirror in the right wedge ($\mathcal{R}$). Reflection focuses to a future sink ($\mathcal{F}$). A similar balancing situation exists in the left wedge.

temperature $T_H = \hbar a/2\pi k_B c$, consistent with the standard identification of the horizon as a thermal object. The Rindler wedge interior, where the mirror operates, is taken as the cold reservoir at temperature T_C. The identification of T_C with the Landauer erasure temperature of the mirror is an ansatz of the present construction; its precise value determines the Carnot efficiency $\eta = 1 - T_C/T_H$.

- **Isothermal Expansion:** The bilocal source prepares an entangled photon pair in the basis that $\mathcal{R}$ measures in (H_+/H_- see section 6). In this basis, the Shannon entropy of the Rindler outcome distribution increases from 0 to 1 bit, reflecting the transition from a deterministic prior state to a maximally uncertain Bell state induced by the bilocal entanglement structure, notionally "drawing some heat from the environment $\mathcal{P}$".
- **Adiabatic Expansion:** This corresponds to the crossing over the past Rindler horizon from hot to cold, the absorption of the photon by the mirror.
- **Isothermal Compression:** This corresponds to controlled erasure of the detector's stored bit during the re-emission of the photon by the mirror. This is where the Landauer cost is paid [13]: an energy on the order of $k_B T_C \ln 2$ is dissipated into the cold reservoir.
- **Adiabatic Compression:** This corresponds to the crossing over the future Rindler horizon from cold to hot, the absorption of the future sink $\mathcal{F}$ and reseting of the state to an eigenstate of H_+ or H_-, zero bits of Shannon entropy.

7.1 Shannon Entropy

The entropy variation in the isothermal legs should be understood as Shannon entropy associated with measurement outcomes in a fixed operational basis, rather than the von Neumann entropy of the global quantum state, which remains pure throughout. From an information-theoretic perspective, aligned with Copenhagen interpretation and QBism, the quantum state encodes an observer's knowledge rather than an objective microstate, and thermodynamic entropy should therefore be identified with the classical uncertainty in measurement outcomes. In this sense, Shannon entropy provides the operationally relevant notion entering Landauer's principle, resolving the apparent tension between unitary evolution (which preserves von Neumann entropy) and thermodynamic irreversibility (which is manifest at the level of classical information processing).

No decoherence assumption is required; Shannon entropy here is defined over the classical outcome distribution induced by a fixed measurement basis (the Rindler detector algebra), independent of the unitary evolution of the global state.

7.2 Work and Efficiency

The thermodynamic state space of the cycle is parameterised by temperature T and Shannon entropy H. The two isothermal legs operate at T_H and T_C respectively, and the entropy change per cycle is exactly one bit:

$$\Delta H = \ln 2 \quad \text{(one binary helicity outcome)}, \tag{43}$$

so the work extracted per cycle is the area of the rectangle in T-H space:

$$W = (T_H - T_C)\,\Delta H = (T_H - T_C)\ln 2. \tag{44}$$

The work done on the mirror by absorbing one photon of energy $\hbar\omega$ is the radiation pressure impulse $\Delta p = \hbar\omega/c$, which is responsible for the mirror's acceleration into the Rindler wedge. This identifies the hot reservoir temperature via

$$k_B T_H = \hbar\omega, \tag{45}$$

the energy of one photon sets the Unruh temperature of the horizon. The Carnot efficiency bound $W \leq T_H \Delta H$ then gives

$$W \leq \hbar\omega \ln 2 = \frac{\hbar a \ln 2}{2\pi c}, \tag{46}$$

where in the last step we used $k_B T_H = \hbar a / 2\pi k_B c$ from the Unruh temperature. This is precisely the Unruh–Landauer bound (3) at $\Delta H = \ln 2$:

$$\boxed{W \leq \frac{\hbar a}{2\pi c}\Delta H \iff \Delta E \geq \frac{\hbar a}{2\pi c}\Delta H.} \tag{47}$$

The Carnot efficiency bound for this cycle and the Unruh–Landauer bound are therefore the same inequality, approached from opposite directions: the Carnot bound is a statement about the maximum work extractable from the thermal gradient between the horizon and the wedge interior, while the Unruh–Landauer bound is a statement about the minimum energy cost of acquiring one bit at that horizon. Their equivalence identifies the Rindler horizon as a Carnot engine operating at the Unruh temperature, with quantum measurement as the working substance and the classical bit as the thermodynamic output.

The cold reservoir temperature T_C is the Landauer temperature associated with isolating the mirror from its local re-emission environment, while T_H is the Landauer temperature associated with isolating the entire Rindler wedge from the global Minkowski vacuum. The two temperatures correspond to two levels of coarse-graining: T_H traces out the opposite wedge at the Rindler horizon, while T_C traces out only the mirror's local photon field at the scale of the mirror interaction. The temperature gradient $T_H > T_C$ is therefore a gradient between two scales of isolation, and the Carnot efficiency $\eta = 1 - T_C/T_H$ measures how much of the global thermal gradient is captured by the local mirror interaction. In the limit $T_C \to 0$ the mirror's local environment is fully decoupled and the cycle extracts maximum work; in the limit $T_C \to T_H$ the two scales of isolation coincide and the cycle is in hydrostatic equilibrium with no net work extracted.

The efficiency $\eta = 1 - T_C/T_H$ is maximised in the limit $T_C \to 0$, which corresponds to perfect erasure at zero thermodynamic cost — the ideal mirror limit. Any residual $T_C > 0$ reflects irreversibility in the mirror's erasure process, consistent with Landauer's principle applied to the re-emission step.

8 Effective Field Theory Structure

The bilocal construction has a natural effective field theory interpretation. The right wedge $\mathcal{R}$ is the *system*, with energy scale $E_{\rm sys}$; the full Minkowski space $\mathcal{R} \cup \mathcal{L}$ is the *environment*, with energy scale $E_{\rm env} = E_{\rm sys} + E_{\mathcal{L}}$. The Rindler horizon is the EFT cutoff surface: it is the boundary at which the system's description and the full Minkowski description become incompatible, and the left wedge $\mathcal{L}$ is precisely the degrees of freedom that the system has integrated out.

8.1 The GHY Term as Effective Action

We now show that the GHY boundary term arises as the effective action for $\mathcal{H}_R$ after integrating out $\mathcal{H}_L$ in the path integral, elevating the identification $S_{GHY}|_{\mathcal{H}} = S_{\rm matter}$ from Section 5 from an algebraic observation to a structural result.

Setup. The full Euclidean path integral for gravity and matter decomposes over the two wedges as

$$Z = \int \mathcal{D}g_R\, \mathcal{D}\phi_R \int \mathcal{D}g_L\, \mathcal{D}\phi_L\; e^{-S_R - S_L - S_{\mathcal{H}}}, \tag{48}$$

where $S_{\mathcal{H}}$ is the boundary term at the horizon coupling the two wedges. Integrating out the left wedge defines the effective action:

$$e^{-S_{\rm eff}[g_R,\phi_R]} = \int \mathcal{D}g_L\, \mathcal{D}\phi_L\; e^{-S_L - S_{\mathcal{H}}}. \tag{49}$$

Origin of the horizon coupling. Integration by parts applied to any kinetic term produces a boundary contribution at $\mathcal{H}$. For the gravitational field this is the Gibbons–Hawking–York term [10]

$$S_{\mathcal{H}} = S_{GHY}\big|_{\mathcal{H}} = \frac{1}{8\pi G} \int_{\mathcal{H}} d^2x_\perp\, d\lambda\, \theta, \tag{50}$$

which couples the two wedge geometries through their expansion θ at the horizon, precisely as a scalar kinetic term couples ϕ_L and ϕ_R through their normal derivatives.

Saddle point evaluation. The classical solution in the left wedge is flat Minkowski space perturbed by the AS pp-wave of the left-moving photon. Aichelburg–Sexl pp-waves are Ricci flat ($R_{\mu\nu} = 0$) in vacuum, so the bulk Einstein–Hilbert action vanishes on-shell:

$$S_L\big|_{\text{on-shell}} = \frac{1}{16\pi G} \int_L d^4x \sqrt{g}\, R = 0. \tag{51}$$

The left wedge contributes only through the horizon coupling and the matter action:

$$\int \mathcal{D}g_L\, \mathcal{D}\phi_L\; e^{-S_L - S_{\mathcal{H}}}\bigg|_{\text{saddle}} = e^{-S_{GHY}|_{\mathcal{H}} - S^L_{\rm matter}}. \tag{52}$$

Since $S_{GHY}|_{\mathcal{H}} = S^L_{\rm matter} = \sum_i \hbar\omega_i$ by equation (42), the effective action for the right wedge is

$$\boxed{S_{\rm eff}[g_R,\phi_R] = S_R[g_R,\phi_R] + S_{GHY}\big|_{\mathcal{H}}.} \tag{53}$$

The GHY term is the contribution to the right wedge effective action from integrating out the left wedge at the classical level. The bulk left wedge action vanishes by Ricci-flatness of the AS solution; the entire left wedge effect is encoded in the boundary term at the horizon.

Quantum corrections. Fluctuations of ϕ_L around the saddle generate corrections to S_{eff} suppressed by ℓ_P^2/A relative to the classical term. These are the standard entanglement entropy contributions studied by Susskind–Uglum [30] and Callan–Wilczek [31], whose UV divergences renormalize Newton's constant G. Equation (53) is the leading classical contribution.

8.2 Partition Function and S-Matrix

The thermal partition function of the right wedge

$$Z = \mathrm{Tr}_{\mathcal{H}_R}\, e^{-H_R/k_B T_H} \tag{54}$$

is obtained from the full Minkowski vacuum by tracing out $\mathcal{H}_L$: integrating out the left wedge in the path integral yields the thermal density matrix for $\mathcal{H}_R$, and $\ln Z$ generates the connected correlation functions of the right wedge EFT. The S-matrix of the full bilocal construction connects the past state at $\mathcal{P}$ to the future sink $\mathcal{F}$:

$$S = \langle \mathcal{F} | \mathcal{P} \rangle = \mathrm{Tr}\, e^{-iHt/\hbar}, \tag{55}$$

which under Wick rotation $\tau = it$ becomes $Z = \mathrm{Tr}\, e^{-H\tau/\hbar}$. The Carnot cycle of Section 7 is therefore the thermodynamic avatar of this S-matrix element: the real-time bilocal construction and its thermodynamic description are related by Wick rotation, and each microstate (23) contributes one term to Z rather than the full thermal trace.

8.3 Horizon Complementarity

The identification of $\mathcal{H}_R$ as system and $\mathcal{H}_L \otimes \mathcal{H}_R$ as environment is the operational content of horizon complementarity [32]: the right wedge observer sees a thermal state at T_H, while the full Minkowski observer sees a pure entangled state. These are the EFT and UV descriptions of the same microstate. The spatial bit — the answer to "who observed H_+?" — is visible to $\mathcal{R}$ as a classical record but its quantum origin in the entangled pair is accessible only to the full Minkowski observer who holds both wedges. Complementarity is not an additional postulate here but an operational consequence of the EFT structure: the horizon is the surface at which the single-wedge and two-wedge descriptions become incompatible, consistently with the equivalence list of the following subsection.

8.4 EFT Validity and the Planck Scale

The construction defines two natural length scales. The system scale is the Compton wavelength of the detector

$$\lambda_C = \frac{\hbar}{m_{\mathcal{R}} c}, \tag{56}$$

the scale below which the single-particle description of $\mathcal{R}$ breaks down. The environment scale is the proper horizon distance

$$d_{\mathcal{H}} = \frac{c^2}{a}, \tag{57}$$

the scale at which the single-bit description saturates. The EFT is valid when these scales are separated, $\lambda_C \ll d_{\mathcal{H}}$; it breaks down when they coincide. The Planck length

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} = \sqrt{\lambda_C \cdot r_s} \tag{58}$$

is the geometric mean of the Compton wavelength and the Schwarzschild radius $r_s = 2Gm/c^2$: it is the unique length scale built from both $\hbar$ and G and is the scale at which $\lambda_C = r_s$, i.e. where the system scale and the gravitational scale coincide. The Planck scale is therefore the fixed point of the EFT: the unique scale at which the quantum and gravitational descriptions become equally valid and the EFT has no regime of applicability.

The Carnot efficiency of Section 7 measures the separation between these scales. With $T_H = \hbar a/2\pi k_B c$ and $T_C = \hbar\kappa/2\pi k_B c$, the efficiency

$$\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{\kappa}{a} = \frac{m_{\mathcal{L}}}{m_{\mathcal{R}} + m_{\mathcal{L}}} \tag{59}$$

is the fraction of the total mass residing in the environment — the degrees of freedom integrated out. The following conditions are simultaneously equivalent:

Proposition 1 (EFT Validity). *The following are equivalent for the bilocal construction:*

1. $\kappa < a$ *(mirror surface gravity less than global Rindler acceleration)*
2. $E_{\text{sys}} < E_{\text{env}}$ *(system energy less than environment)*
3. $\eta > 0$ *(Carnot cycle extracts positive work)*
4. $\lambda_C < d_{\mathcal{H}}$ *(Compton scale inside horizon distance)*
5. $\ell_P/\lambda_C \ll 1$ *(equivalently $m_{\mathcal{R}} \gg m_P$, the detector mass is super-Planckian)*
6. $\mathcal{R}$ *is a proper subsystem of its environment.*

The boundary case where all inequalities become equalities is simultaneously the horizon condition, the Planck condition $\lambda_C = r_s = \ell_P$, and the thermodynamic equilibrium condition $\eta = 0$. The EFT breaks down exactly at the horizon, which is not a failure of the framework but its natural boundary condition.

9 Towards Experimental Tests

The Unruh–Landauer relation (3) is, in principle, an experimentally accessible statement: it bounds the energy cost of acquiring one bit of information by a detector undergoing proper acceleration a, with a specific coefficient $\hbar/2\pi c$ that is not a free parameter. However, at laboratory accelerations the correction to the standard Landauer erasure floor is of order 10^{-44} J per nat, far below the thermal noise floor of current experiments. Direct measurement of the UL correction in a table-top setting is not currently feasible. We outline four directions in which the framework makes contact with experiment at a level closer to reach, ranging from near-term precision measurement to longer-horizon analogue gravity.

Optomechanical back-action scaling. High-Q optomechanical oscillators already operate near the standard quantum limit, where radiation pressure constitutes the dominant back-action on the mirror. The UL relation predicts an additional contribution to the back-action energy per bit extracted, linear in the effective mirror acceleration a_{eff} induced by the radiation field:

$$\Delta E_{\text{back-action}} \geq \Delta E_{\text{SQL}} + \frac{\hbar a_{\text{eff}}}{2\pi c}\,\Delta H. \tag{60}$$

The effective acceleration is controllable via the intracavity photon number, and the prediction is that the back-action floor has a component with the specific linear scaling and coefficient above, distinguishable from the standard quantum limit by its dependence on a_{eff}. This is not a measurement of a small absolute energy but of a scaling law with a controllable parameter, which is a more accessible experimental target. Next-generation optomechanical experiments approaching the Heisenberg limit [33] may be sensitive to corrections of this form.

Saturation wavelength in high-intensity laser–atom interactions. The saturation condition

$$\lambda_{\mathrm{sat}} = \frac{2\pi}{\ln 2} \cdot \frac{c^2}{a} \tag{61}$$

identifies a preferred photon wavelength for single-bit measurement at acceleration a. For electrons in high-intensity laser fields, where effective accelerations can reach $a \sim 10^{25}$ m s^{-2}, the saturation wavelength falls in the nanometre (X-ray) range, which is directly accessible. More structurally, when $a \sim mc^3/\hbar$ (the Compton-scale acceleration), $\lambda_{\mathrm{sat}} \sim \lambda_C/\ln 2$, where $\lambda_C = \hbar/mc$ is the Compton wavelength. The saturation condition is thus approached precisely in the regime where the single-particle approximation breaks down and pair production becomes significant. Whether this coincidence reflects a deeper connection between the UL bound and the onset of quantum field-theoretic effects is an open question, but it suggests that strong-field laser–atom experiments [34] are a natural arena in which to probe the saturation condition.

Weak measurement and irreducible back-action. The framework takes as primitive that every measurement event involves a nonzero energy–momentum transfer. This is a stronger claim than the Heisenberg uncertainty principle in a specific operational sense: weak measurement schemes can in principle extract information with arbitrarily small *recorded* disturbance by displacing the back-action into unmeasured degrees of freedom. The UL framework predicts that this displacement does not eliminate the physical energy transfer; the energy is deposited into the detector's matter-memory regardless of whether it is captured in the measurement record. This distinguishes the present framework from interpretations in which measurement has no special physical content, and from purely relational approaches in which back-action is epistemic rather than physical. The experimental target is not the UL coefficient itself but the *location* of the deposited energy: does a weak measurement that minimises recorded back-action still deposit energy into unmeasured degrees of freedom at the level predicted by the UL relation? Experiments that directly measure the back-action of individual variable-strength quantum measurements, such as [35], provide a natural platform for testing whether the deposited energy scales with the UL coefficient, independently of the measurement record.

Analogue gravity: sonic horizons in Bose–Einstein condensates. Analogue Hawking radiation has been observed in BEC sonic horizon experiments [36], where the role of the speed of light is played by the speed of sound c_s and the effective Unruh temperature is $T_U = \hbar a_{\mathrm{eff}}/2\pi k_B c_s$, with a_{eff} the velocity gradient at the sonic horizon. The UL relation in this analogue setting reads

$$\Delta E \geq \frac{\hbar a_{\mathrm{eff}}}{2\pi c_s} \, \Delta H, \tag{62}$$

with a_{eff} controllable by the condensate parameters. The analogue acceleration can be made large enough that the right-hand side is no longer negligible compared to the energy scales of the system. A detector realized as a localised impurity or probe field near the sonic horizon would, according to the present framework, exhibit an information acquisition cost with this scaling. Identifying the appropriate operational notion of measurement and detector in the condensate setting requires further theoretical development, but the analogue framework is the most promising near-term route to reaching the relevant acceleration regime experimentally.

10 Conclusion

We have identified quantum measurement events as gravitational microstates in the Rindler setting, partially answering the question left open by Jacobson and Verlinde of what physical process underlies each individual entropy increment. The central result is the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$: the gravitational boundary action and the matter action are the same quantity counted in two languages, with the Planck area ℓ_P^2 as the conversion factor. This follows from a single temperature-eliminated Unruh–Landauer bound, from which standard black hole thermodynamics emerges via Newton's laws, and is consistent across thermodynamic, geometric, information-theoretic, and effective field theory descriptions of the horizon, the Carnot efficiency bound and the Unruh–Landauer bound being the same inequality viewed from opposite directions. Matter and horizon are orthogonal information channels: matter the timelike substrate of emergent proper time, the horizon the spacelike surface through which measurement events generate emergent space, evocative of ER=EPR at the single-microstate level. Whether this identification extends from the boundary into the bulk, whether the helicity structure of the emergent space constitutes a topological invariant, and whether the framework admits falsifiable consequences distinguishing it from standard thermodynamic treatments, remain open questions for future work.

11 Appendix: Consistency of the Double-Null Gauge Condition

We verify that the double-null gauge condition $A_+ = A_- = 0$ is consistent with Maxwell's equations and leaves exactly two physical degrees of freedom.

11.1 Coordinates and Notation

In double-null coordinates (x^+, x^-, x^1, x^2) the Minkowski metric reads

$$ds^2 = -2\,dx^+\,dx^- + (dx^1)^2 + (dx^2)^2, \tag{63}$$

so that $\partial_+ = \partial/\partial x^+$, $\partial_- = \partial/\partial x^-$, and the d'Alembertian is

$$\Box = -2\partial_+\partial_- + \partial_i\partial^i, \qquad i = 1, 2. \tag{64}$$

The gauge field has components $A_\mu = (A_+, A_-, A_1, A_2)$ and transforms as

$$A_\mu \to A_\mu + \partial_\mu \Lambda \tag{65}$$

under a gauge transformation with parameter $\Lambda(x^+, x^-, x^1, x^2)$.

11.2 Imposing the Gauge Conditions in Stages

The single gauge parameter Λ is a function of all four coordinates. We use it in three stages, each consuming a different functional dependence.

Stage 1: $A_+ = 0$. Requiring $A_+ + \partial_+\Lambda = 0$ determines Λ as a function of x^+:

$$\Lambda(x^+, x^-, x^i) = -\int^{x^+} A_+(s, x^-, x^i)\,ds + \Lambda_0(x^-, x^i), \tag{66}$$

where $\Lambda_0(x^-, x^i)$ is an arbitrary residual function independent of x^+.

Stage 2: $A_- = 0$. Under the residual freedom $\Lambda_0(x^-, x^i)$, the A_- component transforms as $A_- \to A_- + \partial_-\Lambda_0$. Setting this to zero determines Λ_0 as a function of x^-:

$$\Lambda_0(x^-, x^i) = -\int^{x^-} A_-(s, x^i)\,ds + \Lambda_1(x^i), \tag{67}$$

where $\Lambda_1(x^i)$ is a residual function of the transverse coordinates only.

Stage 3: Transverse Coulomb condition $\partial^i A_i = 0$. Under the remaining freedom $\Lambda_1(x^i)$, the transverse components transform as $A_i \to A_i + \partial_i \Lambda_1$. Imposing $\partial^i A_i = 0$ requires

$$\nabla_\perp^2 \Lambda_1 = -\partial^i A_i, \tag{68}$$

which is a Poisson equation in the transverse plane with a unique solution (up to harmonic functions, fixed by boundary conditions). This exhausts the gauge freedom entirely.

The three conditions have consumed the full gauge parameter Λ in stages, with no over-determination: each stage uses a strictly different functional dependence of Λ.

11.3 Physical Degrees of Freedom

After imposing $A_+ = A_- = 0$ and $\partial^i A_i = 0$, the remaining components are A_1 and A_2 subject to one differential constraint $\partial^i A_i = 0$. In momentum space this reads $k^i A_i = 0$.

For our specific construction, the photons propagate in the $\pm z$ direction, so their wave-vectors are purely null: $k_\perp^i = 0$ for $i = 1, 2$. The transverse Coulomb condition $k^i A_i = 0$ is therefore trivially satisfied for any A_1, A_2, and both transverse components are independent. The two circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \tag{69}$$

are the two independent physical degrees of freedom, corresponding to helicity $\lambda = \pm 1$. This is the correct count for a massless spin-1 field in four dimensions.

The trivial satisfaction of the transverse Coulomb condition relies on the photons propagating purely in the $\pm z$ direction with vanishing transverse momentum $k_\perp^i = 0$. For photons with nonzero transverse momentum the condition $k^i A_i = 0$ imposes a non-trivial constraint and the degree of freedom count requires revisiting.

11.4 Consistency with Maxwell's Equations

With $A_+ = A_- = 0$ and $\partial^i A_i = 0$, the non-zero field strength components are

$$F_{+i} = \partial_+ A_i, \qquad F_{-i} = \partial_- A_i, \qquad F_{ij} = \partial_i A_j - \partial_j A_i, \qquad F_{+-} = 0. \tag{70}$$

Maxwell's equations $\partial^\mu F_{\mu\nu} = J_\nu$ decompose as follows.

Transverse components $\nu = i$:

$$-2\partial_+ \partial_- A_i + \partial^j (\partial_j A_i - \partial_i A_j) = J_i. \tag{71}$$

Using $\partial^i A_i = 0$, the second term simplifies to $\partial^j \partial_j A_i$, giving the standard wave equation

$$\boxed{\Box A_i = J_i.} \tag{72}$$

Null components $\nu = \pm$:

$$-\partial^i \partial_\pm A_i = J_\pm. \tag{73}$$

These are constraint equations relating the null source components to the transverse field. For our bilocal source, which is purely transverse, $J_+ = J_- = 0$, and the constraints reduce to

$$\partial^i \partial_\pm A_i = 0. \tag{74}$$

Differentiating the transverse Coulomb condition $\partial^i A_i = 0$ with respect to $x^\pm$ gives $\partial^i (\partial_\pm A_i) = 0$, so these constraints are automatically satisfied. ✓

11.5 Summary

The double-null gauge condition $A_+ = A_- = 0$ supplemented by the transverse Coulomb condition $\partial^i A_i = 0$:

1. uses the gauge freedom of Λ completely and without over-determination, via a three-stage procedure consuming the x^+-, x^--, and x^i-dependent parts of Λ in sequence;
2. leaves exactly two physical degrees of freedom, the helicity eigenstates $\lambda = \pm 1$, consistent with the known count for a massless spin-1 field;
3. reduces Maxwell's equations to the standard transverse wave equation $\Box A_i = J_i$, with the null constraint equations automatically satisfied for a purely transverse source.

The gauge is therefore internally consistent and does not over-constrain the physical degrees of freedom. The CPT wedge-swap symmetry is preserved throughout: conditions on A_+ and A_- are imposed symmetrically, and neither null direction is privileged over the other.

Acknowledgements

The author used Claude (Anthropic) as an AI assistant for critical discussion, feedback on arguments, and editorial suggestions during the preparation of this manuscript. All scientific content, calculations, and conclusions are the author's own.

References

[1] T. Jacobson, "Thermodynamics of spacetime: the einstein equation of state," *Physical review letters*, vol. 75, no. 7, p. 1260, 1995.

[2] E. Verlinde, "On the origin of gravity and the laws of newton," *Journal of High Energy Physics*, vol. 2011, no. 4, pp. 1–27, 2011.

[3] W. G. Unruh, "Notes on black-hole evaporation," *Physical Review D*, vol. 14, no. 4, p. 870, 1976.

[4] S. W. Hawking, "Particle creation by black holes," *Communications in mathematical physics*, vol. 43, no. 3, pp. 199–220, 1975.

[5] S. A. Fulling, "Nonuniqueness of canonical field quantization in riemannian spacetime," *Physical Review D*, vol. 7, no. 10, p. 2850, 1973.

[6] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a hermitian scalar field," *Journal of Mathematical Physics*, vol. 16, no. 4, pp. 985–1007, 1975.

[7] R. Penrose, "On gravity's role in quantum state reduction," *General relativity and gravitation*, vol. 28, no. 5, pp. 581–600, 1996.

[8] L. Diosi, "A universal master equation for the gravitational violation of quantum mechanics," *Physics letters A*, vol. 120, no. 8, pp. 377–381, 1987.

[9] K. Player, "Driving the unruh response," *arXiv preprint arXiv:2509.16710*, 2025.

[10] G. W. Gibbons and S. W. Hawking, "Action integrals and partition functions in quantum gravity," *Physical Review D*, vol. 15, no. 10, p. 2752, 1977.

[11] T. Dray and G. Hooft, "The gravitational shock wave of a massless particle," *Nuclear physics B*, vol. 253, pp. 173–188, 1985.

[12] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM journal of research and development*, vol. 5, no. 3, pp. 183–191, 1961.

[13] C. H. Bennett, "The thermodynamics of computation—a review," *International Journal of Theoretical Physics*, vol. 21, no. 12, pp. 905–940, 1982.

[14] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D*, vol. 7, pp. 2333–2346, Apr 1973.

[15] C. C. Gerry and P. L. Knight, *Introductory quantum optics.* Cambridge university press, 2023.

[16] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space.* Cambridge: Cambridge University Press, 1984.

[17] H. J. Bremermann, "Optimization through evolution and recombination.," *Self-organizing systems*, pp. 93–106, 1962.

[18] S. Lloyd, "Ultimate physical limits to computation," *Nature*, vol. 406, no. 6799, pp. 1047–1054, 2000.

[19] D. N. Page and W. K. Wootters, "Evolution without evolution: Dynamics described by stationary observables," *Physical Review D*, vol. 27, no. 12, p. 2885, 1983.

[20] V. Giovannetti, S. Lloyd, and L. Maccone, "Quantum time," *Physical Review D*, vol. 92, no. 4, p. 045033, 2015.

[21] J. A. Wheeler, "Geons," *Physical Review*, vol. 97, no. 2, p. 511, 1955.

[22] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," *Physical review letters*, vol. 59, no. 5, p. 521, 1987.

[23] K. Brown, "Stochastic sprinkling." MathPages. Accessed: 2026-04-19.

[24] R. Penrose, *The road to reality.* Random house, 2006.

[25] L. H. Ryder, *Quantum field theory.* Cambridge university press, 1996.

[26] S. W. Hawking, M. J. Perry, and A. Strominger, "Soft hair on black holes," *Physical Review Letters*, vol. 116, no. 23, p. 231301, 2016.

[27] S. Carlip, "Black hole entropy from bondi-metzner-sachs symmetry at the horizon," *Physical review letters*, vol. 120, no. 10, p. 101301, 2018.

[28] T. Padmanabhan, "Thermodynamical aspects of gravity: new insights," *Reports on Progress in Physics*, vol. 73, no. 4, p. 046901, 2010.

[29] J. Maldacena and L. Susskind, "Cool horizons for entangled black holes," *Fortschritte der Physik*, vol. 61, no. 9, pp. 781–811, 2013.

[30] L. Susskind and J. Uglum, "Black hole entropy in canonical quantum gravity and superstring theory," *Physical Review D*, vol. 50, no. 4, p. 2700, 1994.

[31] C. Callan and F. Wilczek, "On geometric entropy," *Physics Letters B*, vol. 333, no. 1-2, pp. 55–61, 1994.

[32] L. Susskind, L. Thorlacius, and J. Uglum, "The stretched horizon and black hole complementarity," *Physical Review D*, vol. 48, no. 8, p. 3743, 1993.

[33] M. Aspelmeyer, T. J. Kippenberg, and F. Marquardt, "Cavity optomechanics," *Reviews of Modern Physics*, vol. 86, no. 4, pp. 1391–1452, 2014.

[34] A. Di Piazza, C. Müller, K. Hatsagortsyan, and C. H. Keitel, "Extremely high-intensity laser interactions with fundamental¡? format?¿ quantum systems," *Reviews of Modern Physics*, vol. 84, no. 3, pp. 1177–1228, 2012.

[35] M. Hatridge, S. Shankar, M. Mirrahimi, F. Schackert, K. Geerlings, T. Brecht, K. Sliwa, B. Abdo, L. Frunzio, S. M. Girvin, *et al.*, "Quantum back-action of an individual variable-strength measurement," *Science*, vol. 339, no. 6116, pp. 178–181, 2013.

[36] J. Steinhauer, "Observation of quantum hawking radiation and its entanglement in an analogue black hole," *Nature Physics*, vol. 12, no. 10, pp. 959–965, 2016.